

4 General Product Characteristics

4.1 Absolute Maximum Ratings

Table 1 Absolute Maximum Ratings¹⁾

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to ground, positive current flowing into pin
(unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Voltages							
Boost driver ground	V_{BSG}	-0.3	–	0.3	V	–	P_4.1.1
Input standby LDO	V_{VST}	-0.3	–	40	V	2) 3)	P_4.1.2
Input voltage pin 1 (pre regulator)	V_{VS1}	-0.3	–	40	V	2) 3)	P_4.1.3
Input voltage pin 2 (pre regulator)	V_{VS2}	-0.3	–	40	V	2) 3) PG-LQFP-64 only	P_4.1.4
External step up power stage, gate	V_{DRG}	-0.3	–	40	V	2) 3)	P_4.1.5
External power stage, sense resistor high	V_{RSH}	-0.3	–	40	V	2) 3)	P_4.1.6
External power stage, sense resistor low	V_{RSL}	-0.3	–	6.0	V	–	P_4.1.7
Enable	V_{ENA}	-0.3	–	40	V	2) 3)	P_4.1.8
Enable	I_{ENA}	-5	–	–	mA	4)	P_4.1.9
Wake/Inhibit	V_{WAK}	-0.3	–	40	V	2) 3)	P_4.1.10
Wake/Inhibit	I_{WAK}	-5	–	–	mA	4)	P_4.1.11
Reset output	V_{ROT}	-0.3	–	6.0	V	–	P_4.1.12
SPI: Chip select CS	V_{SCS}	-0.3	–	6.0	V	–	P_4.1.13
SPI: Clock CLK	V_{SCL}	-0.3	–	6.0	V	–	P_4.1.14
SPI: Data_In DI	V_{SDI}	-0.3	–	6.0	V	–	P_4.1.15
SPI: Data_Out DO	V_{SDO}	-0.3	–	6.0	V	–	P_4.1.16
Interrupt	V_{INT}	-0.3	–	6.0	V	–	P_4.1.17
Window watchdog trigger	V_{WDI}	-0.3	–	6.0	V	–	P_4.1.18
Error pin	V_{ERR}	-0.3	–	6.0	V	–	P_4.1.19
Safe state 1	V_{SS1}	-0.3	–	6.0	V	–	P_4.1.20
Safe state 2	V_{SS2}	-0.3	–	6.0	V	–	P_4.1.21
Output voltage reference	V_{QVR}	-0.3	–	6.0	V	–	P_4.1.22
Output tracker 2	V_{QT2}	-0.3	–	40	V	–	P_4.1.23
Sense Pin for tracker 2	V_{SQT2}	-0.3	–	40	V	PG-LQFP-64 only	P_4.1.24
Output tracker 1	V_{QT1}	-0.3	–	40	V	–	P_4.1.25

Table 1 Absolute Maximum Ratings¹⁾ (cont'd)

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to ground, positive current flowing into pin
(unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Sense Pin for tracker 1	V_{SQT1}	-0.3	–	40	V	PG-LQFP-64 only	P_4.1.26
Output LDO_Com	V_{QCO}	-0.3	–	6.0	V	–	P_4.1.27
Output LDO_μC	V_{QUC}	-0.3	–	6.0	V	–	P_4.1.28
Sense Pin for LDO_μC	V_{SQUC}	-0.3	–	6.0	V	PG-LQFP-64 only	P_4.1.29
V_Core_Mon ext core supply	V_{VCI}	-0.3	–	6.0	V	–	P_4.1.30
Select ext core supply	V_{SEC}	-0.3	–	6.0	V	–	P_4.1.31
Sync_Out ext core supply	V_{SYN}	-0.3	–	6.0	V	–	P_4.1.32
Enable ext core supply	V_{EVC}	-0.3	–	6.0	V	–	P_4.1.33
FB_BUCK4	V_{FB4}	-0.3	–	7.0	V	PG-LQFP-64 only	P_4.1.35
FB_BUCK3	V_{FB3}	-0.3	–	7.0	V	PG-LQFP-64 only	P_4.1.36
FB_BUCK2	V_{FB2}	-0.3	–	7.0	V	–	P_4.1.37
FB_BUCK1	V_{FB1}	-0.3	–	7.0	V	–	P_4.1.38
BU_GND2	V_{PG2}	-0.3	–	0.3	V	–	P_4.1.39
BU_GND1	V_{PG1}	-0.3	–	0.3	V	–	P_4.1.40
SW2	V_{SW2}	-0.3	–	40	V	PG-LQFP-64 only	P_4.1.41
SW1	V_{SW1}	-0.3	–	40	V	–	P_4.1.42
Select step up pre regulator	V_{STU}	-0.3	–	6.0	V	–	P_4.1.43
FRE	V_{FRE}	-0.3	–	6.0	V	–	P_4.1.44
Q_STBY	V_{QST}	-0.3	–	6.0	V	–	P_4.1.45
MPS	V_{MPS}	-0.3	–	6.0	V	–	P_4.1.46

Temperatures

Junction Temperature	T_j	-40	–	150	°C	–	P_4.1.47
Storage Temperature	T_{stg}	-55	–	150	°C	–	P_4.1.48

ESD Susceptibility

ESD Susceptibility to GND	V_{ESD}	-2	–	2	kV	HBM ⁵⁾	P_4.1.49
ESD Susceptibility to GND	V_{ESD}	-500	–	500	V	CDM ⁶⁾	P_4.1.50
ESD Susceptibility (corner pins) to GND	$V_{\text{ESD,Corner}}$	-750	–	750	V	CDM ⁶⁾	P_4.1.51

1) Not subject to production test, specified by design.

2) Maximum rating is 60 V, if rise time from 0 to 60 V is longer than 10 ms

3) Maximum rating is 49 V, for an overall time of 10 s (in the range of 40 V to 49 V) during the lifetime of the product independent from the rise time.

4) Consider external series resistor for negative voltages < -0.3 V to ensure maximum rating of current

5) ESD susceptibility, HBM according to JEDEC HBM Human Body Model ANSI/ESDA/JEDEC JS001 (1.5k Ω, 100 pF)

6) ESD susceptibility, Charged Device Model “CDM” ESDA STM5.3.1 or ANSI/ESD S.5.3.1

Notes

1. *Stresses above the ones listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.*
2. *Integrated protection functions are designed to prevent IC destruction under fault conditions described in the data sheet. Fault conditions are considered as “outside” normal operating range. Protection functions are not designed for continuous repetitive operation.*

4.2 Functional Range

Note: Within the functional or operating range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given in the Electrical Characteristics table.

Table 2 Functional Range

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Supply Voltage Range for Normal Operation at input of the step-up converter	V_{Bat}	3	–	40	V	with step up pre regulator active in front of step down pre regulator ¹⁾	P_4.2.1
Supply Voltage Range for Normal Operation at pin VSx	V_{VS}	6	–	40	V	without step up pre regulator active in front of step down pre regulator	P_4.2.2
Junction Temperature	T_{j}	-40	–	150	°C	–	P_4.2.3

1) To start up the TLF35584 needs a minimum input voltage of 6.0V at the pin VSx.

Note: Within the functional or operating range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given in the Electrical Characteristics table.

4.3 Thermal Resistance

Note: This thermal data was generated in accordance with JEDEC JESD51 standards. For more information, go to www.jedec.org.

Table 3 Thermal Resistance ¹⁾

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Junction to Case	R_{thJC}	–	–	12.2	K/W	–	P_4.3.1
Junction to Soldering Point (pin)	R_{thJSP}	20.1	–	22.1	K/W	JEDEC 2s2p, measured to pin: VQFN: 1, 6, 7, 28 ; LQFP: 2, 11, 35, 50	P_4.3.2
Junction to Soldering Point (pin)	R_{thJSP}	34.9	–	37.6	K/W	JEDEC 1s0p, measured to pin: VQFN: 1, 6, 7, 28 ; LQFP: 2, 11, 35, 50	P_4.3.3
Junction to Soldering Point (soldering pad)	R_{thJSP}	11.0	–	14.7	K/W	JEDEC 2s2p	P_4.3.4
Junction to Soldering Point (soldering pad)	R_{thJSP}	13.1	–	18.0	K/W	JEDEC 1s0p	P_4.3.5
Junction to Ambient	R_{thJA}	–	37	–	K/W	²⁾	P_4.3.6

1) Not subject to production test, specified by design.

2) Specified R_{thJA} value is according to Jedec JESD51-2,-5,-7 at natural convection on FR4 2s2p board; The Product (Chip+Package) was simulated on a 76.2 × 114.3 × 1.5 mm board with 2 inner copper layers (2 × 70 mm Cu, 2 × 35 mm Cu). Where applicable a thermal via array under the exposed pad contacted the first inner copper layer.

4.4 Quiescent Current Consumption

Table 4 Quiescent current consumption ¹⁾

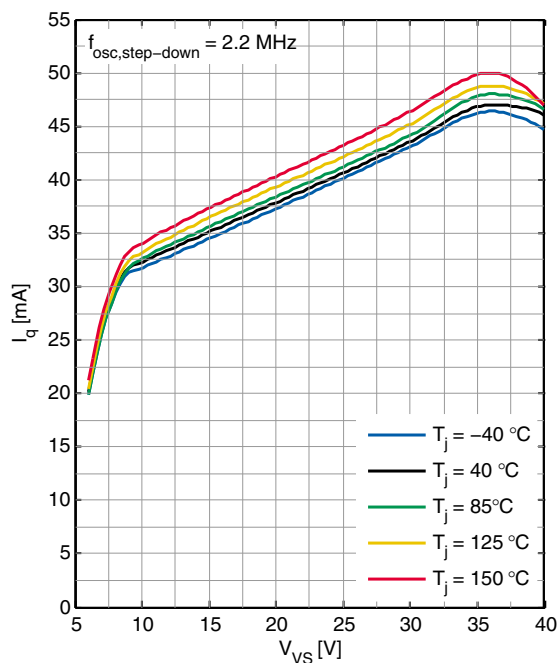
V_{VS} = 6.0 V to 40 V, T_j = -40°C to +150°C, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
INIT state	I_q	—	—	45	mA	¹⁾ $T_j \leq 85^\circ\text{C}$ Step-Up Converter is off, $f_{\text{PREREG, BUCK}} = 2.2 \text{ MHz}$	P_4.4.1
NORMAL state	I_q	—	—	45	mA	¹⁾ $T_j \leq 85^\circ\text{C}$ Step-Up Converter is off, $f_{\text{PREREG, BUCK}} = 2.2 \text{ MHz}$	P_4.4.2
STANDBY state	I_q	—	—	70	μA	¹⁾ LDO_STBY is off $V_{VS} = 14 \text{ V}$; $T_j \leq 40^\circ\text{C}$	P_4.4.3
STANDBY state	I_q	—	—	90	μA	LDO_STBY is off ¹⁾ $T_j \leq 85^\circ\text{C}$	P_4.4.4
STANDBY state	I_q	—	—	120	μA	LDO_STBY is on ¹⁾ $T_j \leq 85^\circ\text{C}$	P_4.4.5
SLEEP state	I_q	—	—	200	μA	¹⁾ $T_j \leq 85^\circ\text{C}$	P_4.4.6
WAKE state	I_q	—	—	45	mA	¹⁾ $T_j \leq 85^\circ\text{C}$ Step-Up Converter is off, $f_{\text{PREREG, BUCK}} = 2.2 \text{ MHz}$	P_4.4.7
FAILSAFE state	I_q	—	—	200	μA	¹⁾ $T_j \leq 85^\circ\text{C}$; $t_{\text{FAILSAFE}} > t_{\text{FAILSAFE, min}}$	P_4.4.8

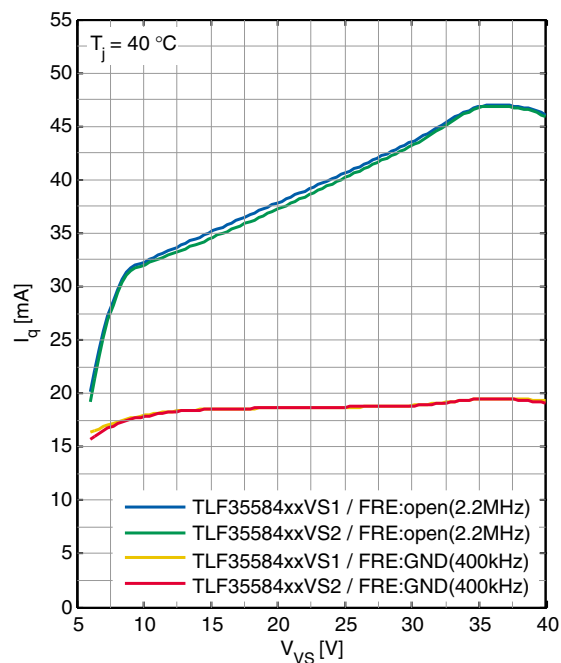
1) All quiescent current parameters are measured at $T_j \leq 85^\circ\text{C}$ and $10 \text{ V} \leq V_{VS} \leq 28 \text{ V}$ with zero load and all selectable options (Outputs, Watchdog, Timers, Step-Up converter) switched off.

4.4.1 Typical Performance Characteristics

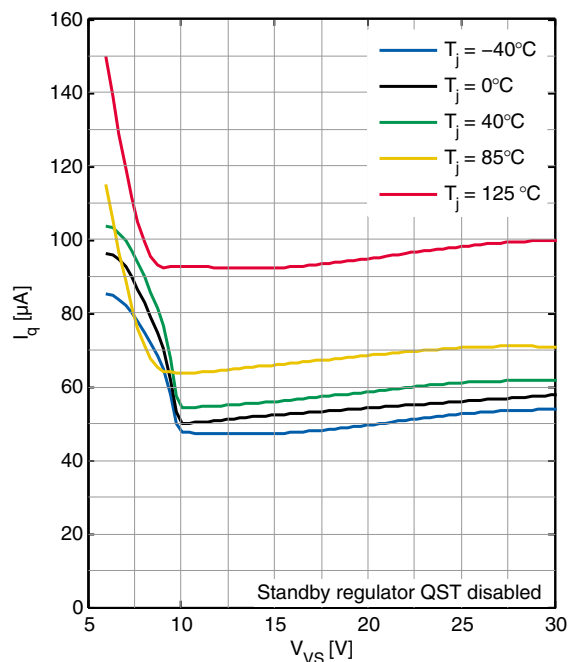
INIT, NORMAL and WAKE Current Consumption I_q versus Supply Voltage V_{VS} (FRE-Pin: open)



INIT, NORMAL and WAKE Current Consumption I_q versus Supply Voltage V_{VS} (various configuration)



STANDBY Current Consumption I_q versus Supply Voltage V_{VS} (QST disabled)



STANDBY Current Consumption I_q versus Supply Voltage V_{VS} (QST enabled)

